

Multiple Choice

1. **Answer:** A
Options: A) gamma rays; B) microwaves; C) ultraviolet radiation; D) visible light
Note: Correct option: A - gamma rays
2. **Answer:** D
Options: A) 0.25 W; B) 0.5 W; C) 1 W; D) 4 W
Note: Correct option: D - 4 W
3. **Answer:** B
Options: A) 0.50 c; B) 0.60 c; C) 0.70 c; D) 0.80 c
Note: Correct option: B - 0.60 c
4. **Answer:** A
Options: A) a dye laser; B) a helium-neon laser; C) an excimer laser; D) a ruby laser
Note: Correct option: A - a dye laser
5. **Answer:** C
Options: A) 1.2×10^9 m/s; B) 3.0×10^8 m/s; C) 1.5×10^8 m/s; D) 1.0×10^8 m/s
Note: Correct option: C - 1.5×10^8 m/s

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'a dye laser'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Planck's constant'.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'tetrahedron'.

Long Form Response

11. **Answer:** The Pauli exclusion principle. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** decreased by a factor of 9. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.